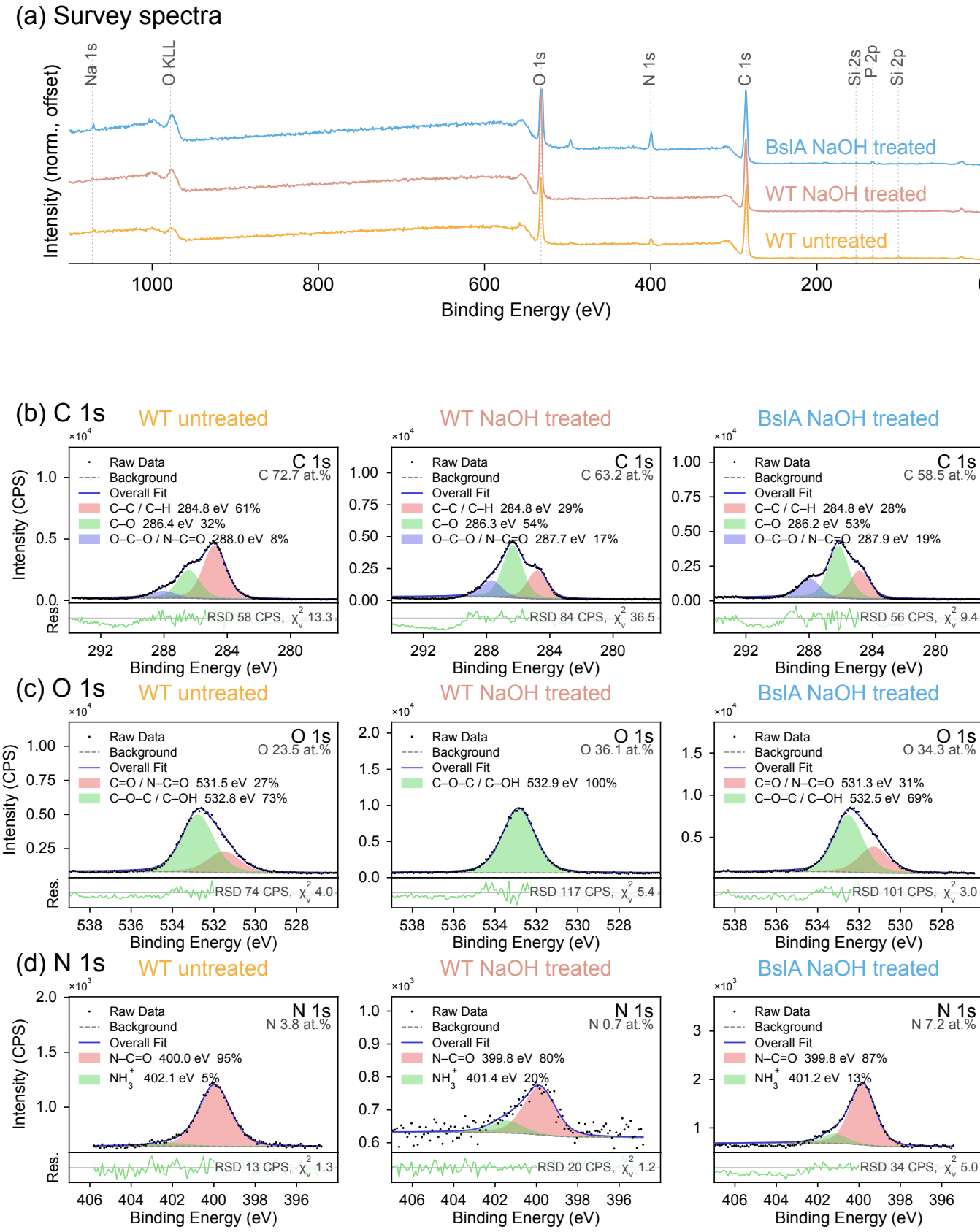

(a) Survey spectra
Na 1s
O KLL
O 1s
N 1s
C 1s
Si 2s
P 2p
Si 2p
Intensity (norm., offset)
BslA NaOH treated
WT NaOH treated
WT untreated
1000
800
600
400
200
0
Binding Energy (eV)
(b) C 1s
WT untreated
WT NaOH treated
BslA NaOH treated
Raw Data
Background
Overall Fit
C 1s
C 72.7 at.%
C–C / C–H 284.8 eV 61%
C–O 286.4 eV 32%
O–C–O / N–C=O 288.0 eV 8%
RSD 58 CPS, χ²ν 13.3
C 63.2 at.%
C–C / C–H 284.8 eV 29%
C–O 286.3 eV 54%
O–C–O / N–C=O 287.7 eV 17%
RSD 84 CPS, χ²ν 36.5
C 58.5 at.%
C–C / C–H 284.8 eV 28%
C–O 286.2 eV 53%
O–C–O / N–C=O 287.9 eV 19%
RSD 56 CPS, χ²ν 9.4
Intensity (CPS)
Res.
Binding Energy (eV)
(c) O 1s
WT untreated
WT NaOH treated
BslA NaOH treated
O 1s
O 23.5 at.%
C=O / N–C=O 531.5 eV 27%
C–O–C / C–OH 532.8 eV 73%
RSD 74 CPS, χ²ν 4.0
O 36.1 at.%
C–O–C / C–OH 532.9 eV 100%
RSD 117 CPS, χ²ν 5.4
O 34.3 at.%
C=O / N–C=O 531.3 eV 31%
C–O–C / C–OH 532.5 eV 69%
RSD 101 CPS, χ²ν 3.0
(d) N 1s
WT untreated
WT NaOH treated
BslA NaOH treated
N 1s
N 3.8 at.%
N–C=O 400.0 eV 95%
NH3+ 402.1 eV 5%
RSD 13 CPS, χ²ν 1.3
N 0.7 at.%
N–C=O 399.8 eV 80%
NH3+ 401.4 eV 20%
RSD 20 CPS, χ²ν 1.2
N 7.2 at.%
N–C=O 399.8 eV 87%
NH3+ 401.2 eV 13%
RSD 34 CPS, χ²ν 5.0